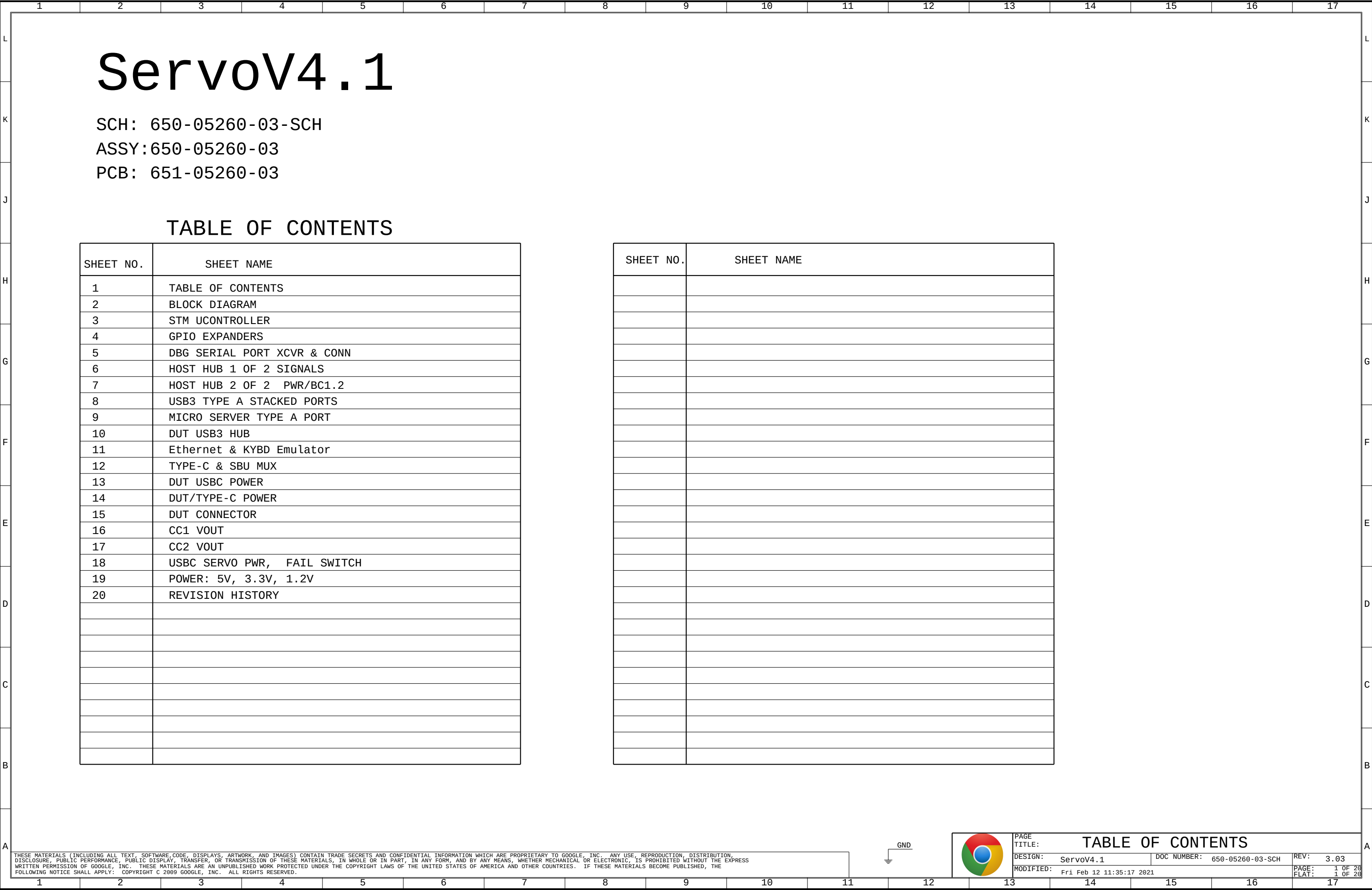
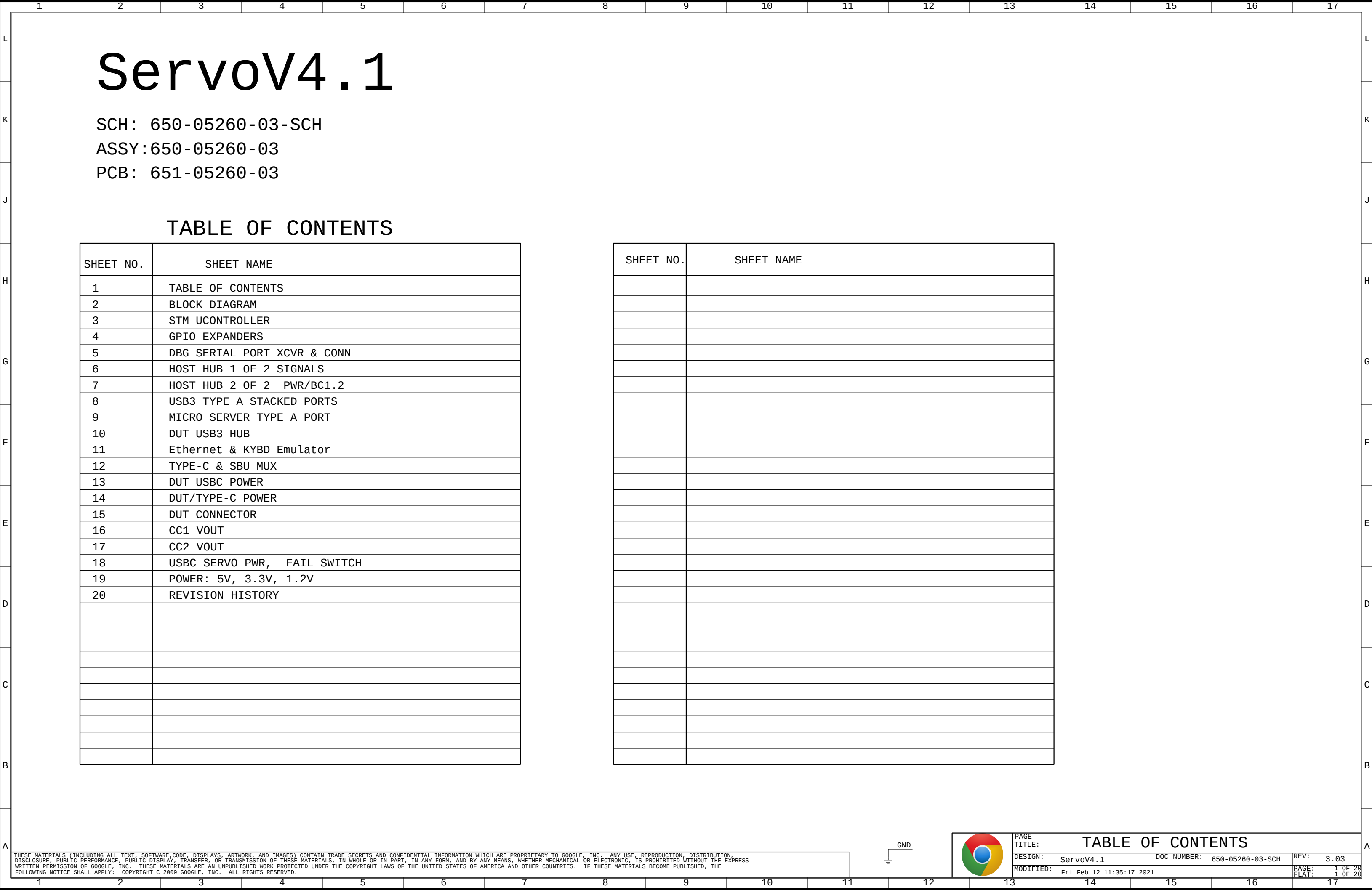
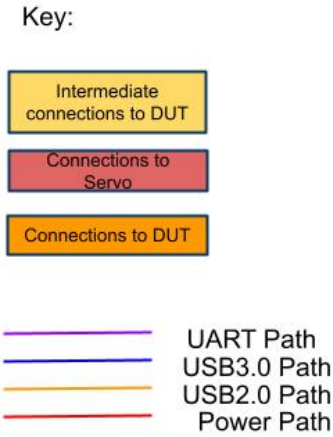
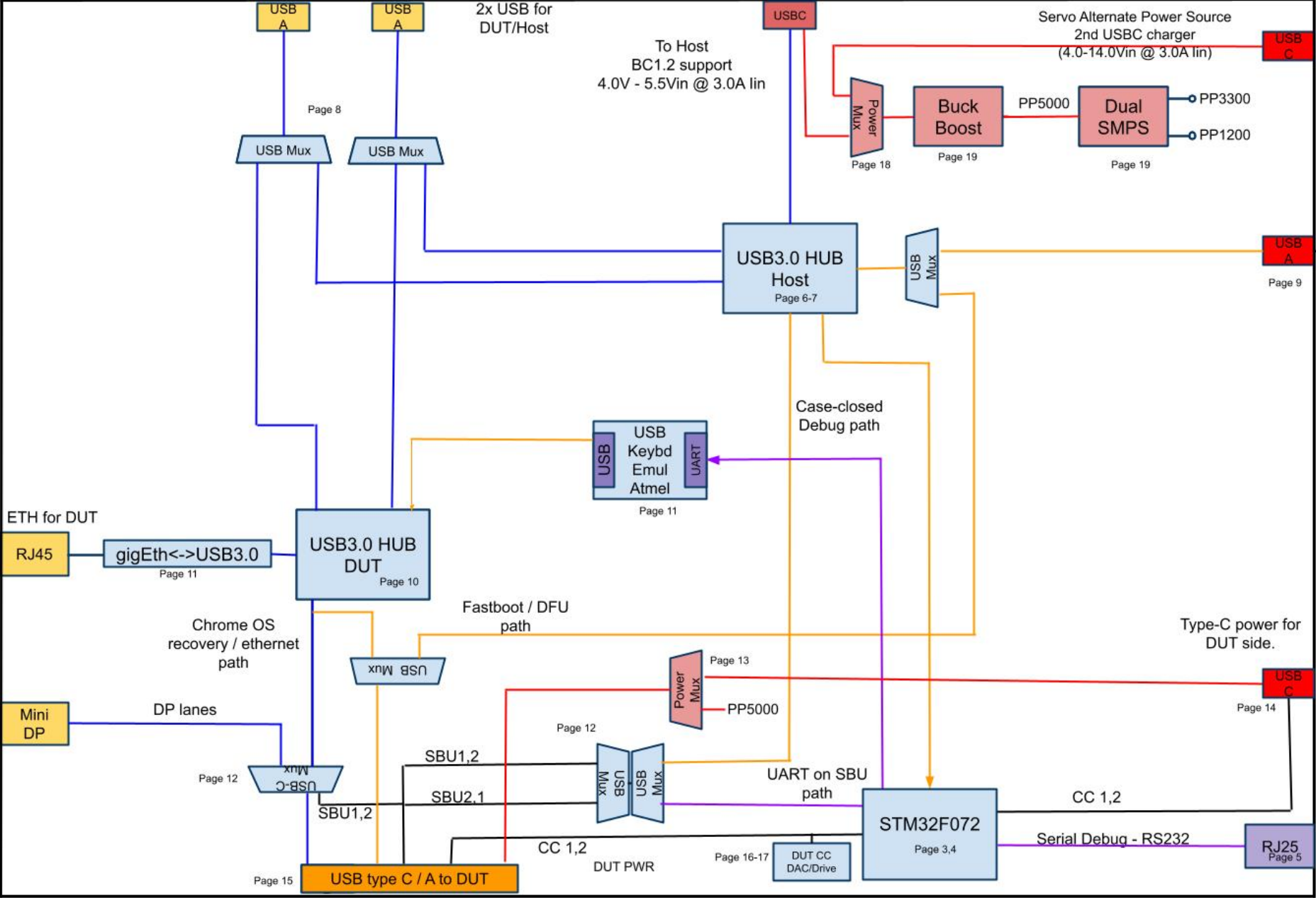


[illegible][illegible][illegible][illegible][illegible]

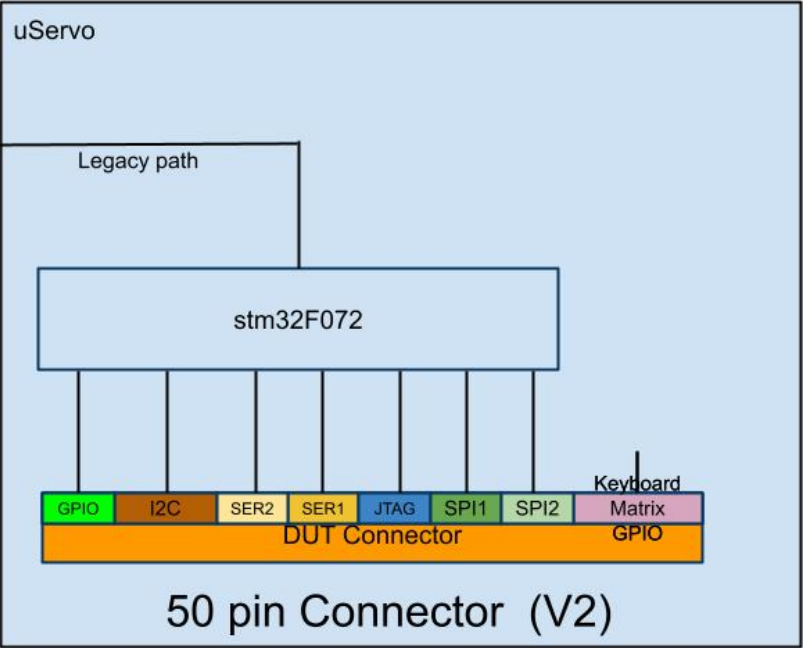
Servo v4.1 Block Diagram
Rev 1.01
2021 Feb 01



DUT USB:
C-C captive cable: for Type-C DUT with CCD
C-A captive cable: for existing Type-A DUT (and use "Legacy path" for "CCD")



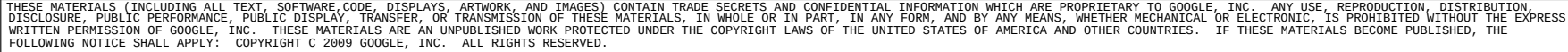
Runs CC/PD code to the DUT enabling host/device and debug options. Allow for DP Alt Mode (can't do with debug on SBU)
Does power negotiation with adapter and pass-through
Could provide any local GPIOs
Could provide UART to USB Kbd emulator and UART on SBU

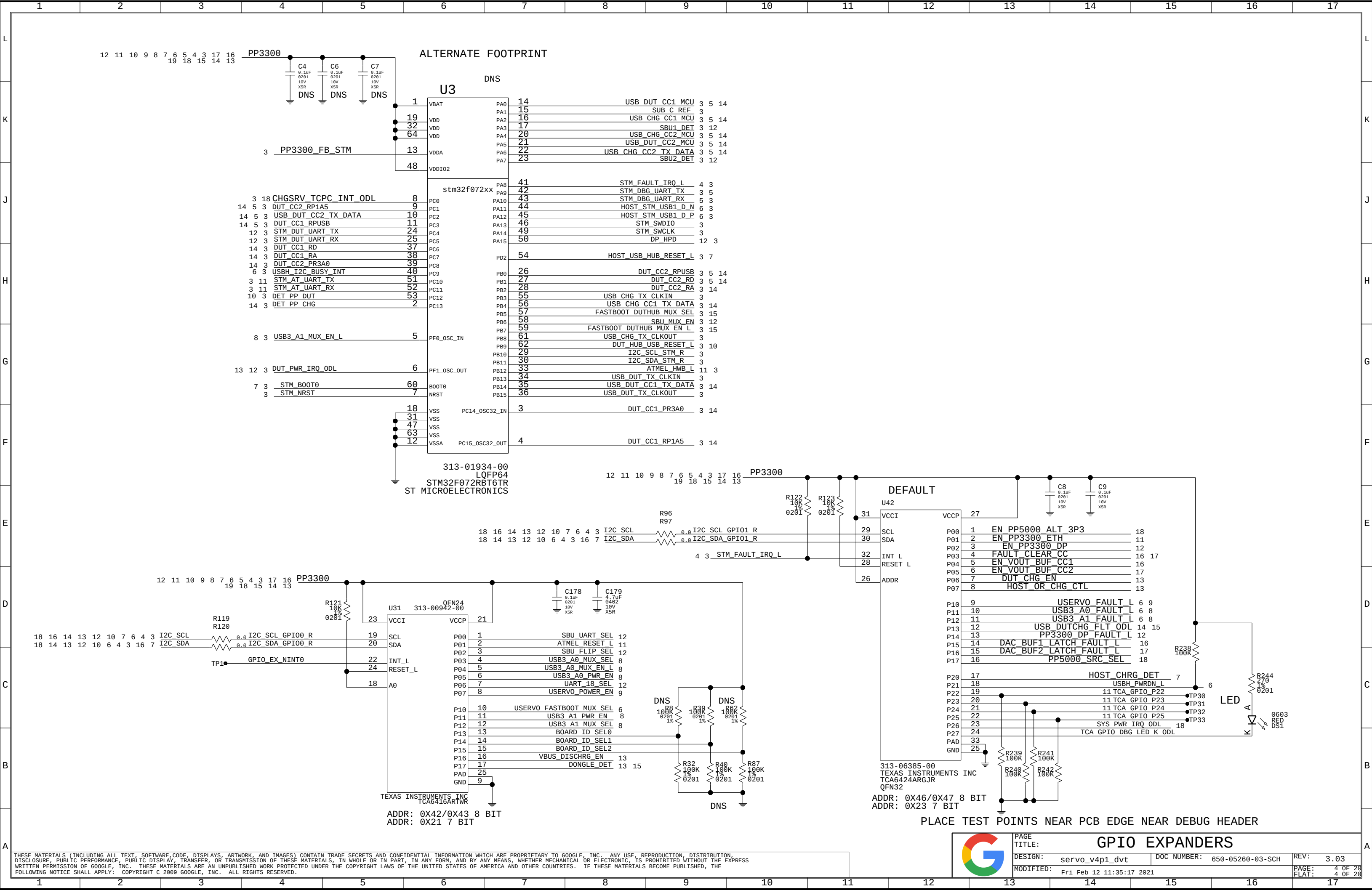


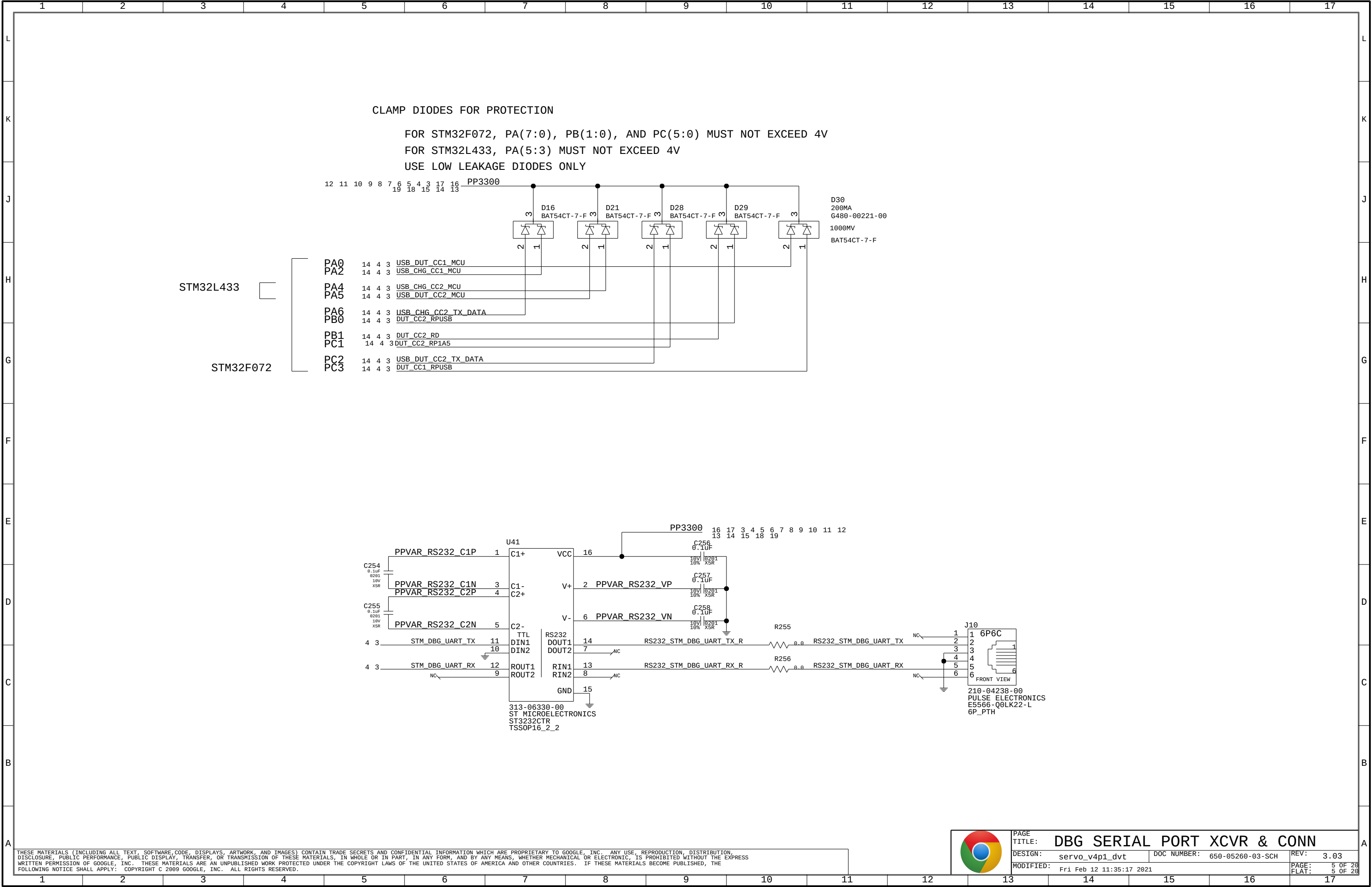
I2C MAP, STM IS MASTER @ 400 KHZ

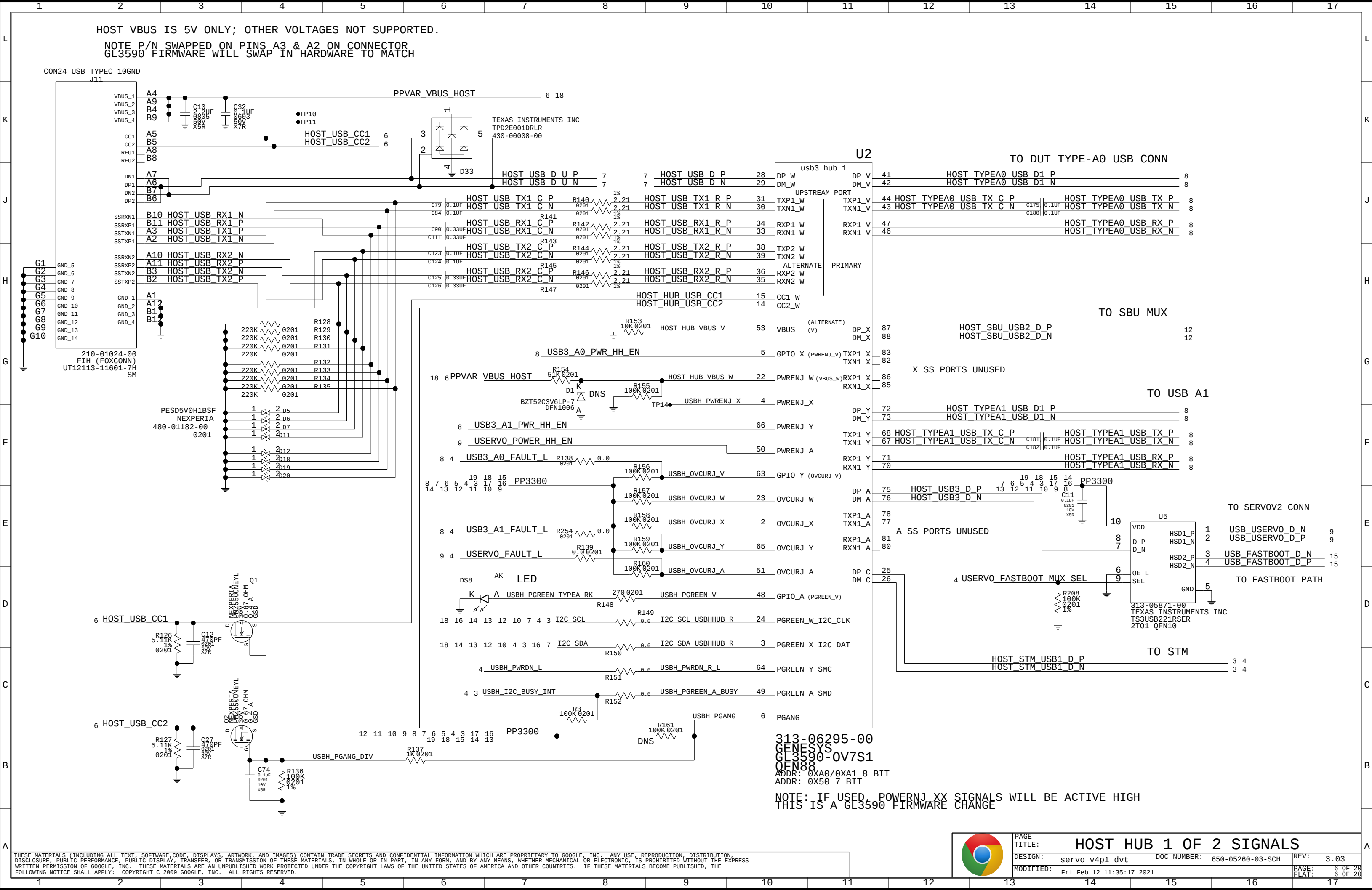
DEVICE	PART	7 BITS	8 BITS READ
USBC/DP REDRIVER	TI TUSB1064	0010 010	0X24
GPIO EXPANDER0	TI TCA6416A	0100 001	0X42
USBC PD CC SENSOR	ON SEMI FUSB302	0100 001	0X44
GPIO EXPANDER1	TI TCA6424A	0100 001	0X46
DUT CURRENT SENSE	TI INA231A	1000 000	0X80
DUT CHR CURRENT SNS	TI INA231A	1000 001	0X82
SRVCHR CURRENT SNS	TI INA231A	1000 010	0X84
CC1 DAC	TI DAC70501	1000 000	0X90
CC2 DAC	TI DAC70501	1000 001	0X92
HOST USB3 HUB	GL GL3590	1010 000	0XA0
BC1.2 DETECTOR	DIODES PI3USB9201	1011 111	0XBE
DUT USB3 HUB	CYPRESS CYUSB3304	1100 000	0XC0

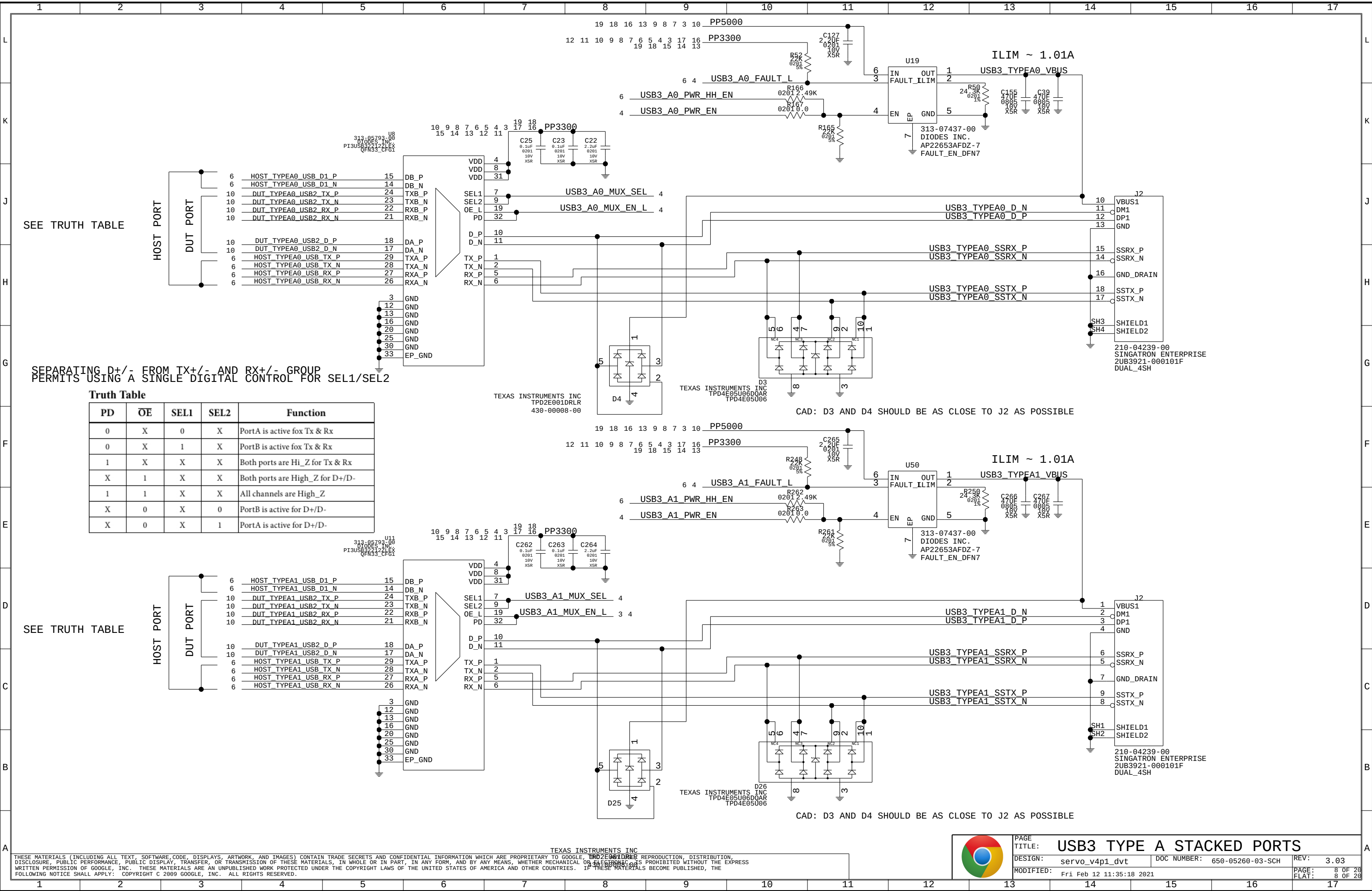












SEE TRUTH TABLE

SEPARATING D+/- FROM TX+/- AND RX+/- GROUP PERMITS USING A SINGLE DIGITAL CONTROL FOR SEL1/SEL2

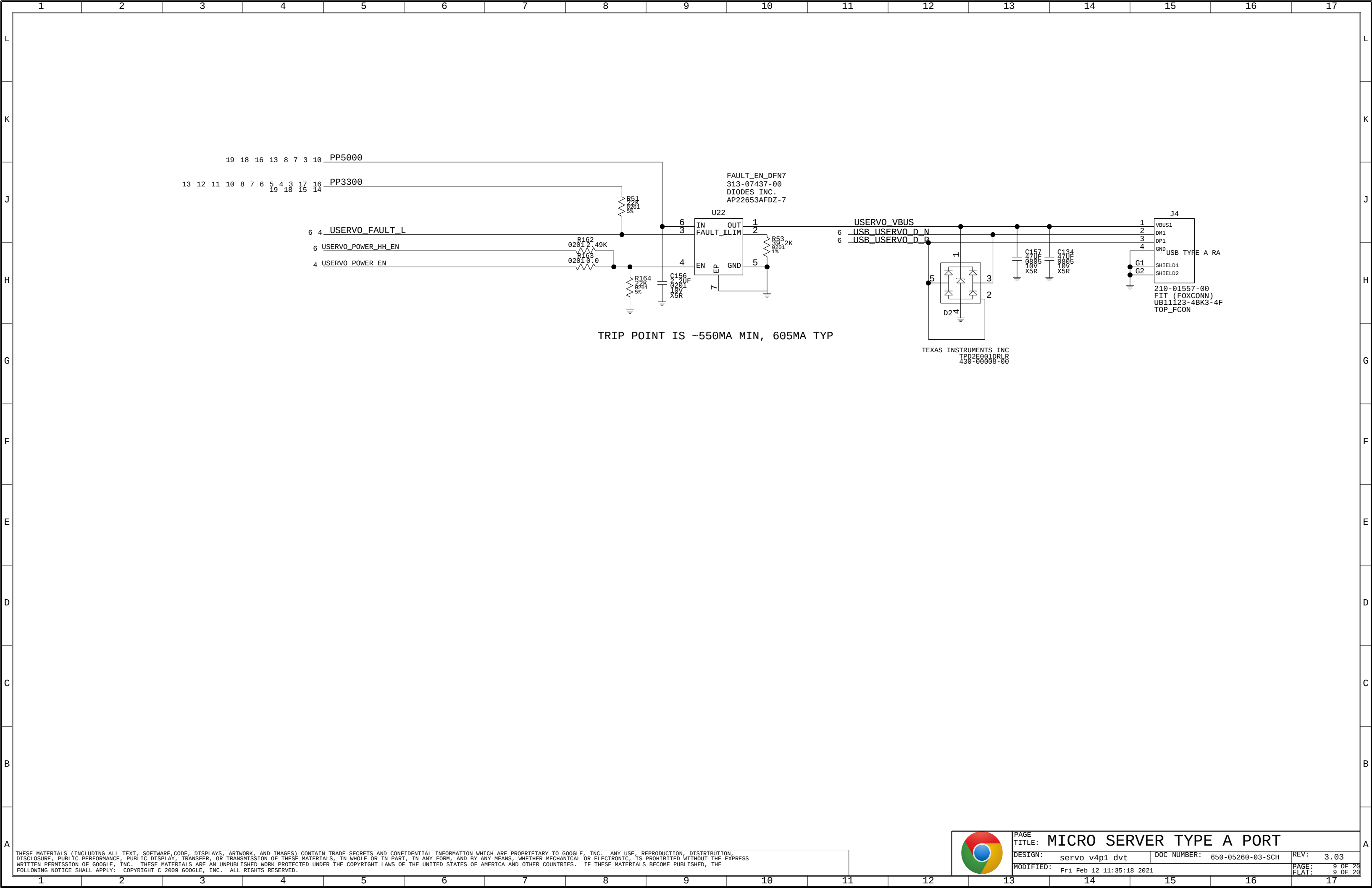
Truth Table

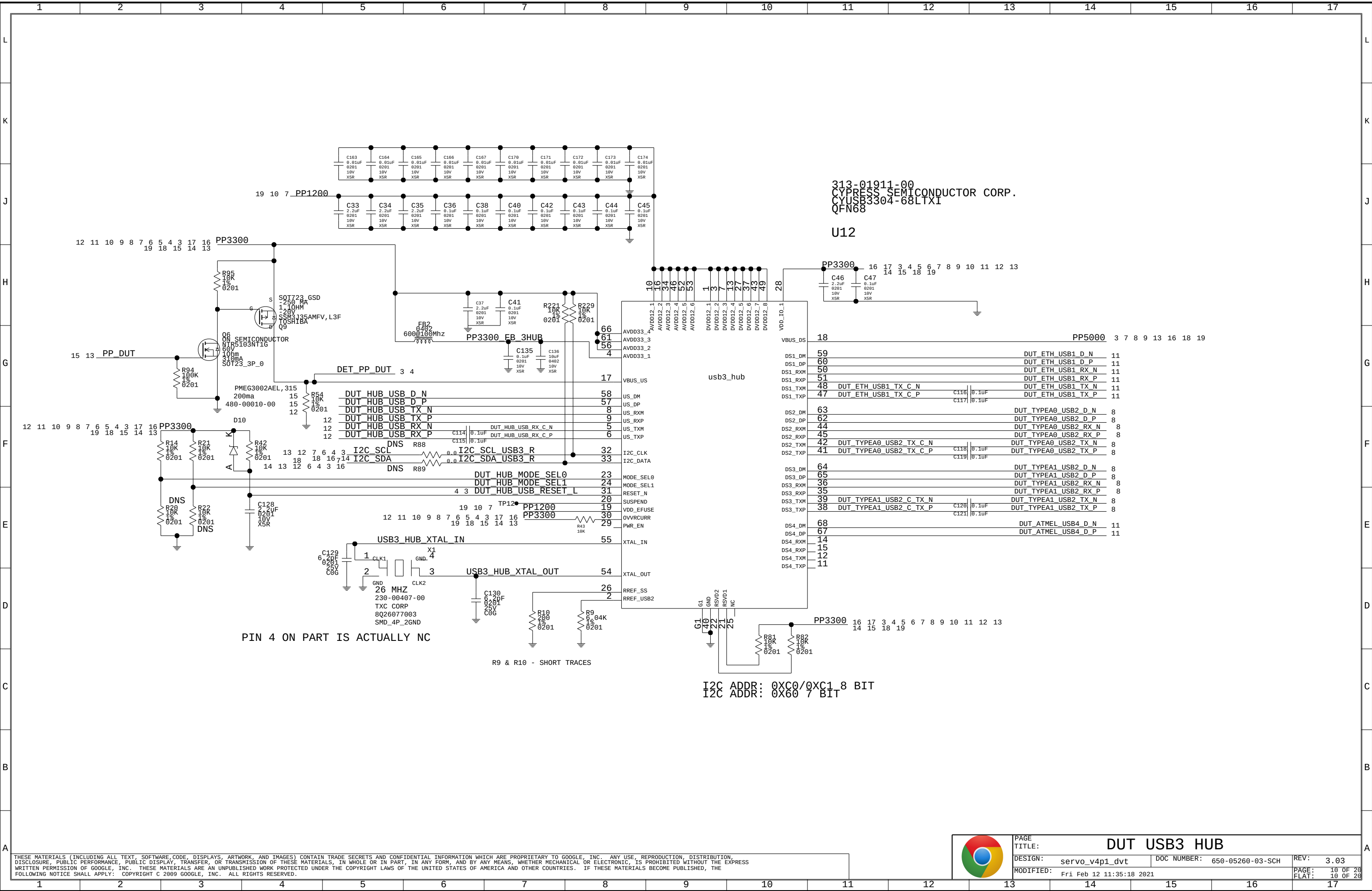
PD	OE	SEL1	SEL2	Function
0	X	0	X	PortA is active fox Tx & Rx
0	X	1	X	PortB is active fox Tx & Rx
1	X	X	X	Both ports are Hi_Z for Tx & Rx
X	1	X	X	Both ports are High_Z for D+/D-
1	1	X	X	All channels are High_Z
X	0	X	0	PortB is active for D+/D-
X	0	X	1	PortA is active for D+/D-

SEE TRUTH TABLE

SEE TRUTH TABLE





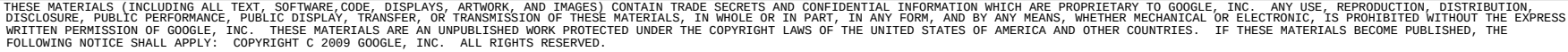


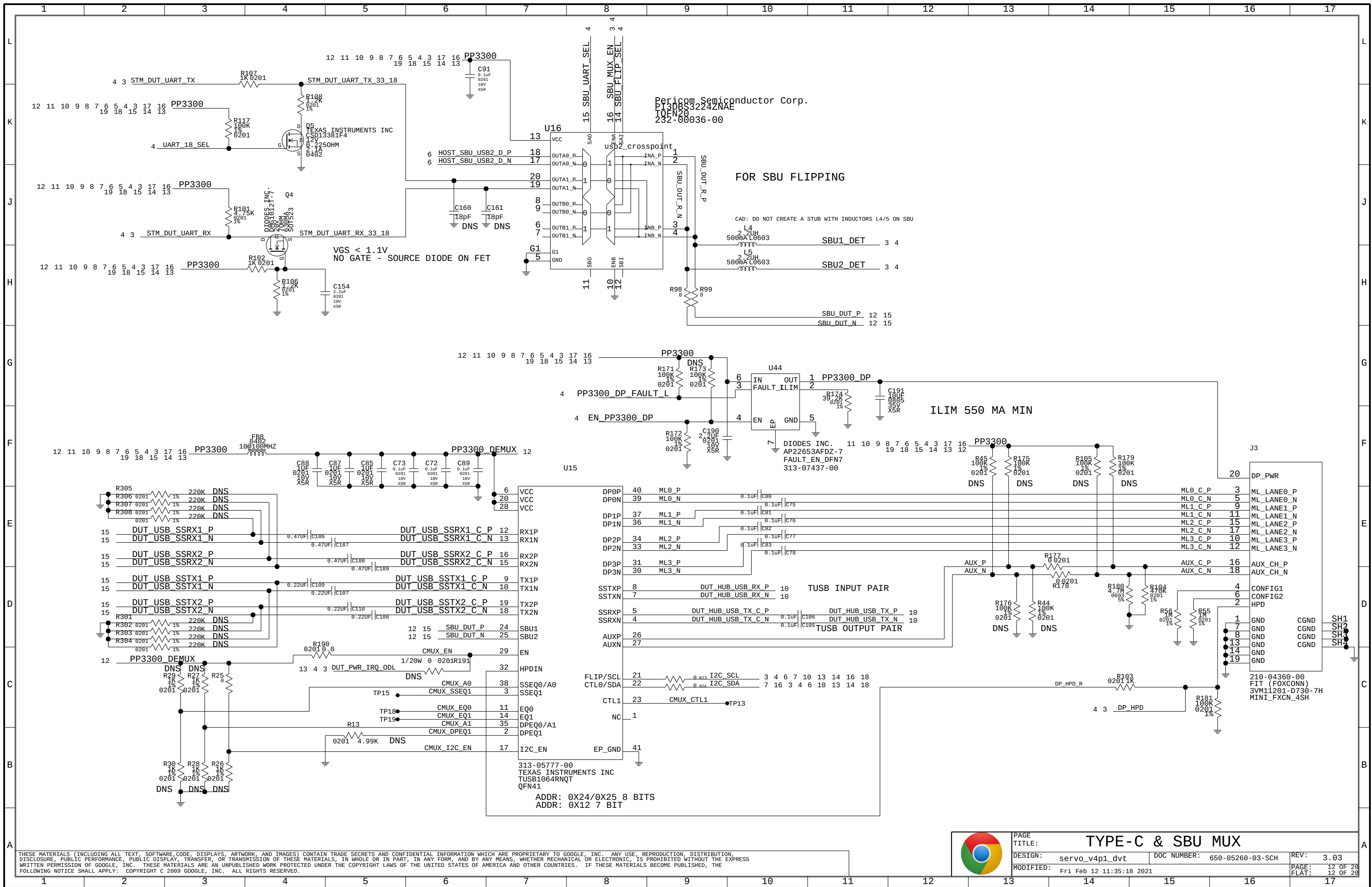
313-01911-00
CYPRESS SEMICONDUCTOR CORP.
CYUSB3304-68LTXI
QFN68

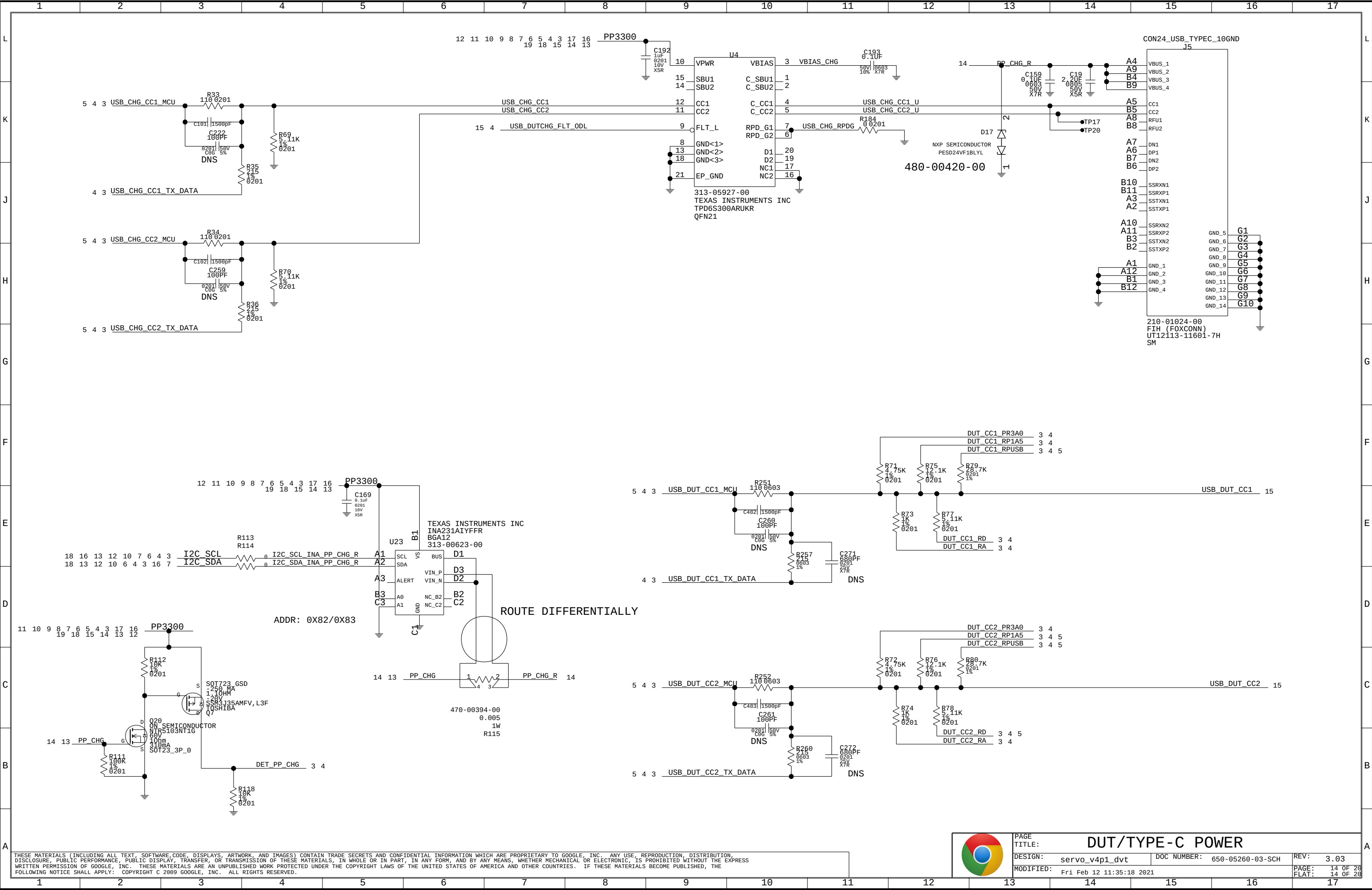
U12

18	PP5000	3	7	8	9	13	16	18	19
59	DUT_ETH_USB1_D_N	11							
60	DUT_ETH_USB1_D_P	11							
50	DUT_ETH_USB1_RX_N	11							
51	DUT_ETH_USB1_RX_P	11							
48	DUT_ETH_USB1_TX_C_N	11							
47	DUT_ETH_USB1_TX_C_P	11							
63	DUT_TYPEA0_USB2_D_N	8							
62	DUT_TYPEA0_USB2_D_P	8							
44	DUT_TYPEA0_USB2_RX_N	8							
45	DUT_TYPEA0_USB2_RX_P	8							
42	DUT_TYPEA0_USB2_TX_C_N	8							
41	DUT_TYPEA0_USB2_TX_C_P	8							
64	DUT_TYPEA1_USB2_D_N	8							
65	DUT_TYPEA1_USB2_D_P	8							
36	DUT_TYPEA1_USB2_RX_N	8							
35	DUT_TYPEA1_USB2_RX_P	8							
39	DUT_TYPEA1_USB2_C_TX_N	8							
38	DUT_TYPEA1_USB2_C_TX_P	8							
68	DUT_ATMEL_USB4_D_N	11							
67	DUT_ATMEL_USB4_D_P	11							

I2C ADDR: 0XC0/0XC1 8 BIT
I2C ADDR: 0X60 7 BIT



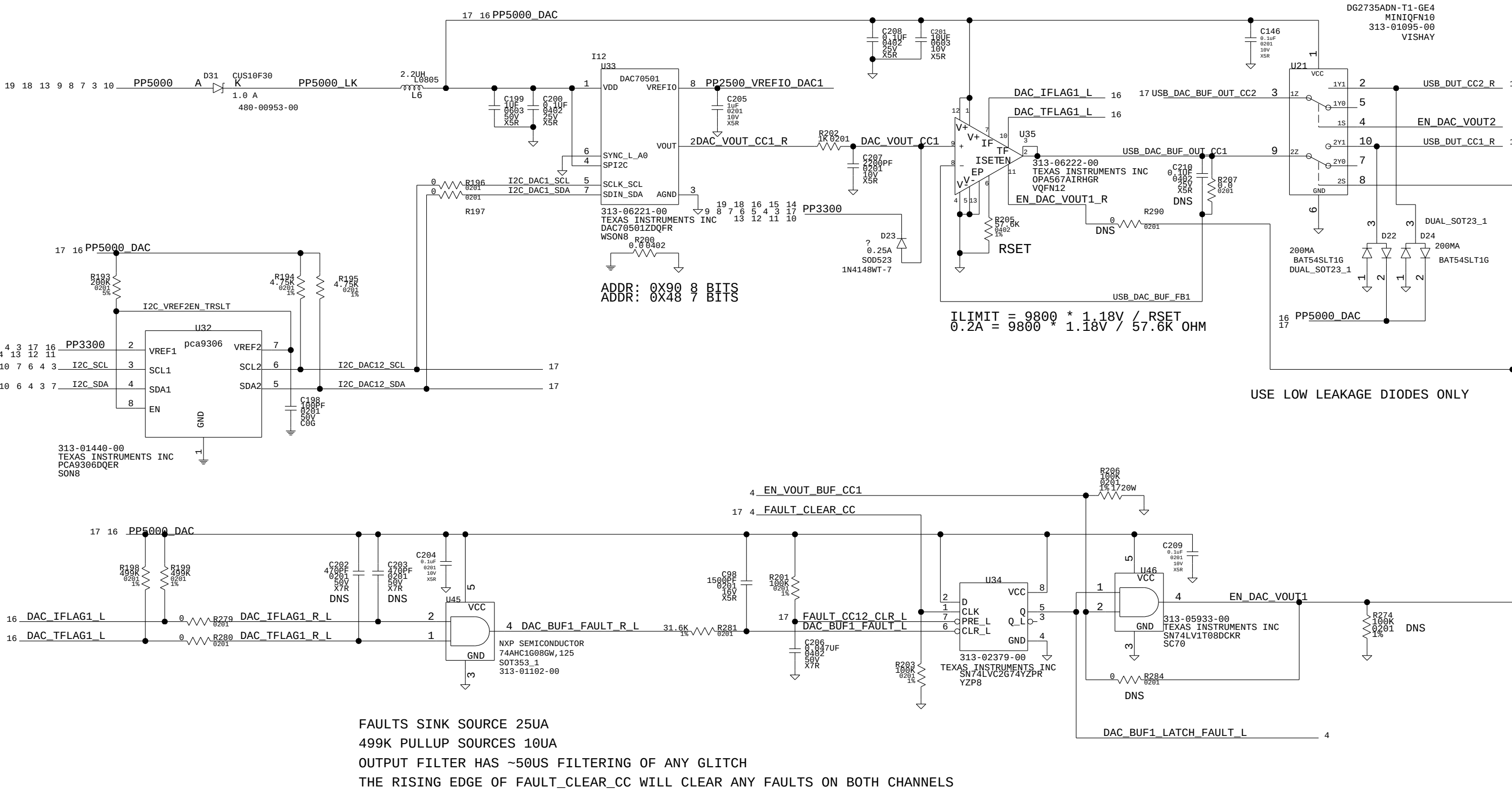


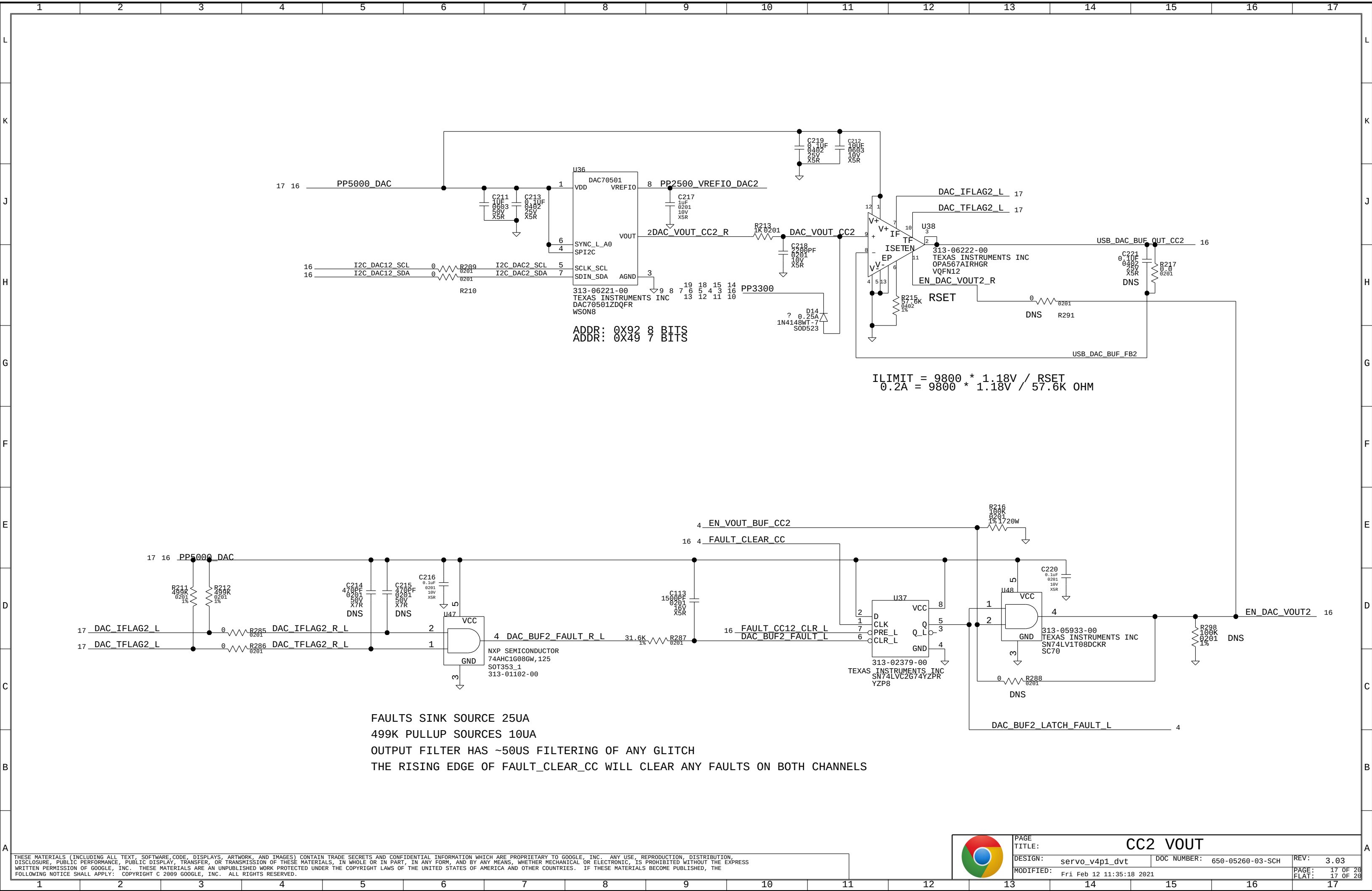


PROGRAMMING NOTES:

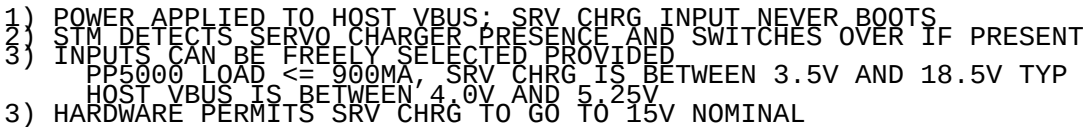
- 1) WITH DAC BUFFERS DISABLED (EN_DAC_VOUT(2:1) = LOW) SET DAC GAIN
- 2) SET DAC OUTPUT VOLTAGE
- 3) ENABLE EN_DAC_VOUT(2:1)

DAC BUFFERS MUST BE DISABLED TO PERMIT CC_PD TRAFFIC
DAC VOLTAGE SHOULD ALWAYS BE SET TO TARGET BEFORE ENABLING BUFFERS!!





WORKAROUND MAY BE TO USE A DUMB EXTERNAL DC SUPPLY WITH A 2.1MM/5.5MM TO USB-C ADAPTER OR SIMILAR CONVERTER.



PAGE				A	
TITLE: USBC SERVO PWR, FAIL SWITCH					
DESIGN: servo_v4p1_dvt		DOC NUMBER: 650-05260-03-SCH		REV: 3.03	
MODIFIED: Tue Mar 16 18:23:33 2021				PAGE: 18 OF 20 FLAT: 18 OF 20	

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